



plum

Behavioural Interventions

to

Reduce No-shows & Late
Cancellations For Teleconsultations

#OnTimeForHealth



Including a
premortem of

5 ways
they'll fail
and how to
prevalidate
to avoid them



Risita Sutar

Outline

Assumptions:

1. 85% of bookings for same day, **largest fraction <1 hr**
2. Cancellations (14%) mean '**Late Cancellations**' for simplicity, and excludes doctor side cancellations
3. 70% of cancellations for Self, rest 30% for dependents
4. Non-attendance cost largest during **peak hours**
5. Cancellation 30 mins early is favourable to **refilling slot** at peak hours
6. Plum wants to build a valuable service which would invite opt-in/add-on purchases by employees (future)

Research:

1. Data from Plum AMA
2. Mental health use tops the usage after GP's
3. Competitors - Even, Practo Instant, Lybrate
4. Interviews with 3 users of telehealth apps
5. Research literature on paid OPD non-attendance
 - a. Data varies greatly, by demography and ailment
In-person: 15-20%
Tele-consult: 6-20%
 - b. Most telehealth bookings fall into - minor GP cases (70%), chronic case follow-up (10%), or referral seekers(20%)

Meet Our Actors

Poh

"The common cold never slowed me down, why should my doctor's appointment"

- I love making the best use of my time, watch me optimize this spreadsheet
- I have trouble saying no to people's face



Neil

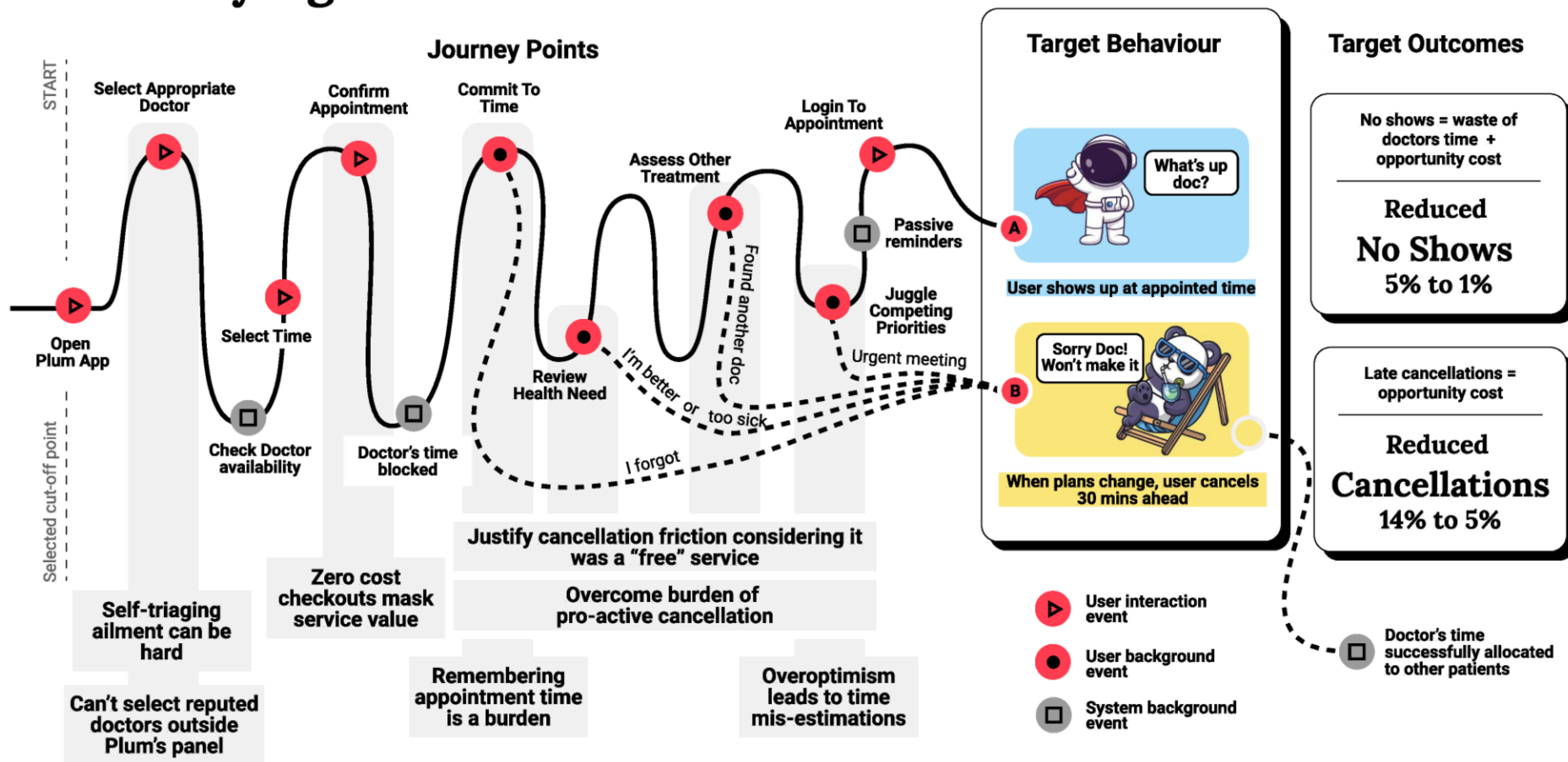
"Sparing time for visits to the doctors can be hard when you have 2 toddlers who are germ magnets"

- I have too much on my plate, things often get lost or delayed
- I'm exhausted admitting mistakes; I often postpone facing the music



As-is Service Blueprint

Identifying outcomes and behaviours



Problem Identification

Barriers contributing to the problem



In telehealth, the **severity of cases** is low or **referral seeking**. Priority is shifted from best treatment to speed and convenience fitting into a work schedule



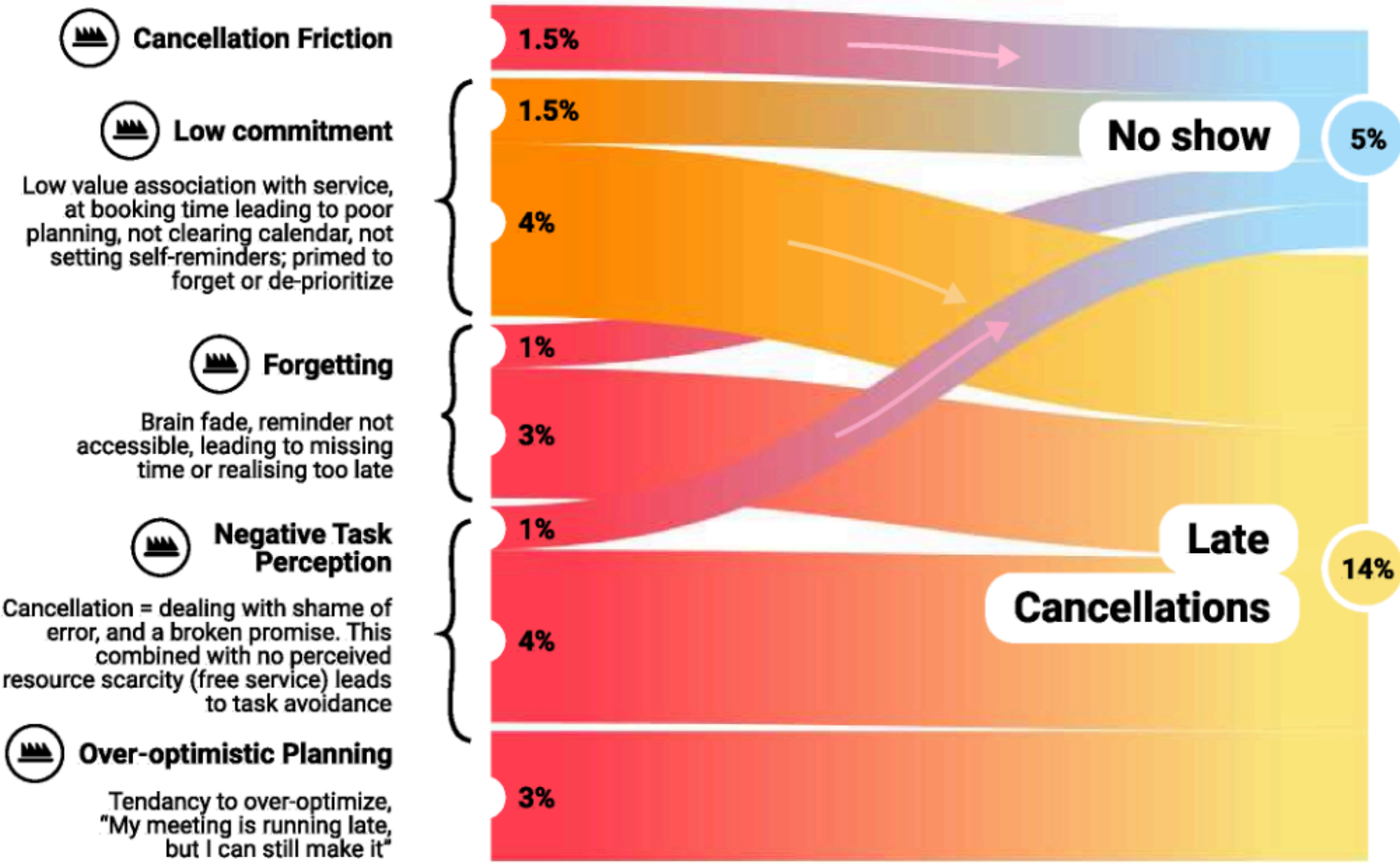
In Plum's case, "free" service removes price barrier, but creates perception of **low value**, on top of the sense of **abundance** created by on-tap "instant" service

What's the worst that could happen?

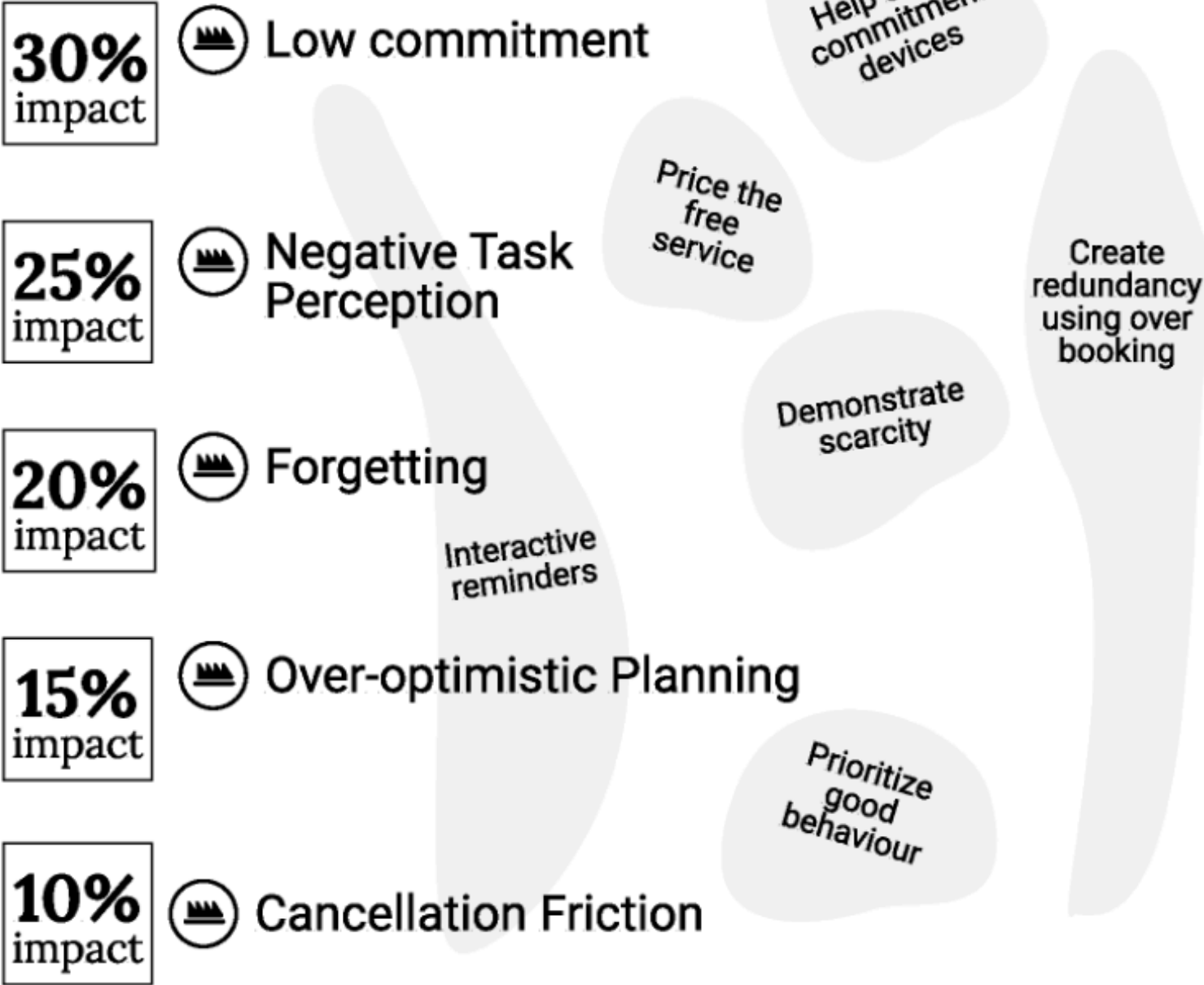


Barriers Contributing to...

... Current Behaviour



Overall Impact and Opportunities



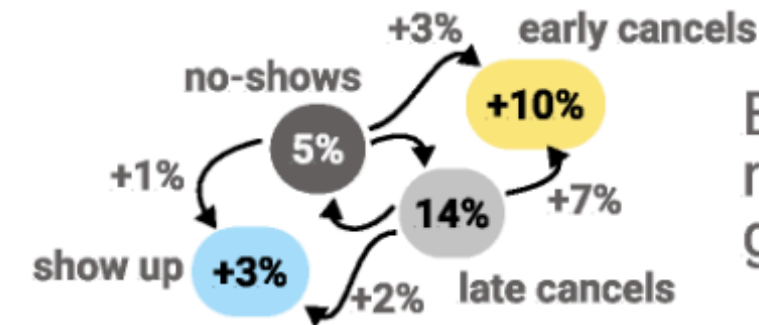
Source for contribution rates: Research literature on medical outpatient non-attendance rates, along with study of relevant cognitive biases, and data provided by Plum

Future Blueprint Ideating and prioritizing interventions

13 Intervention candidates, 9 viable for testing

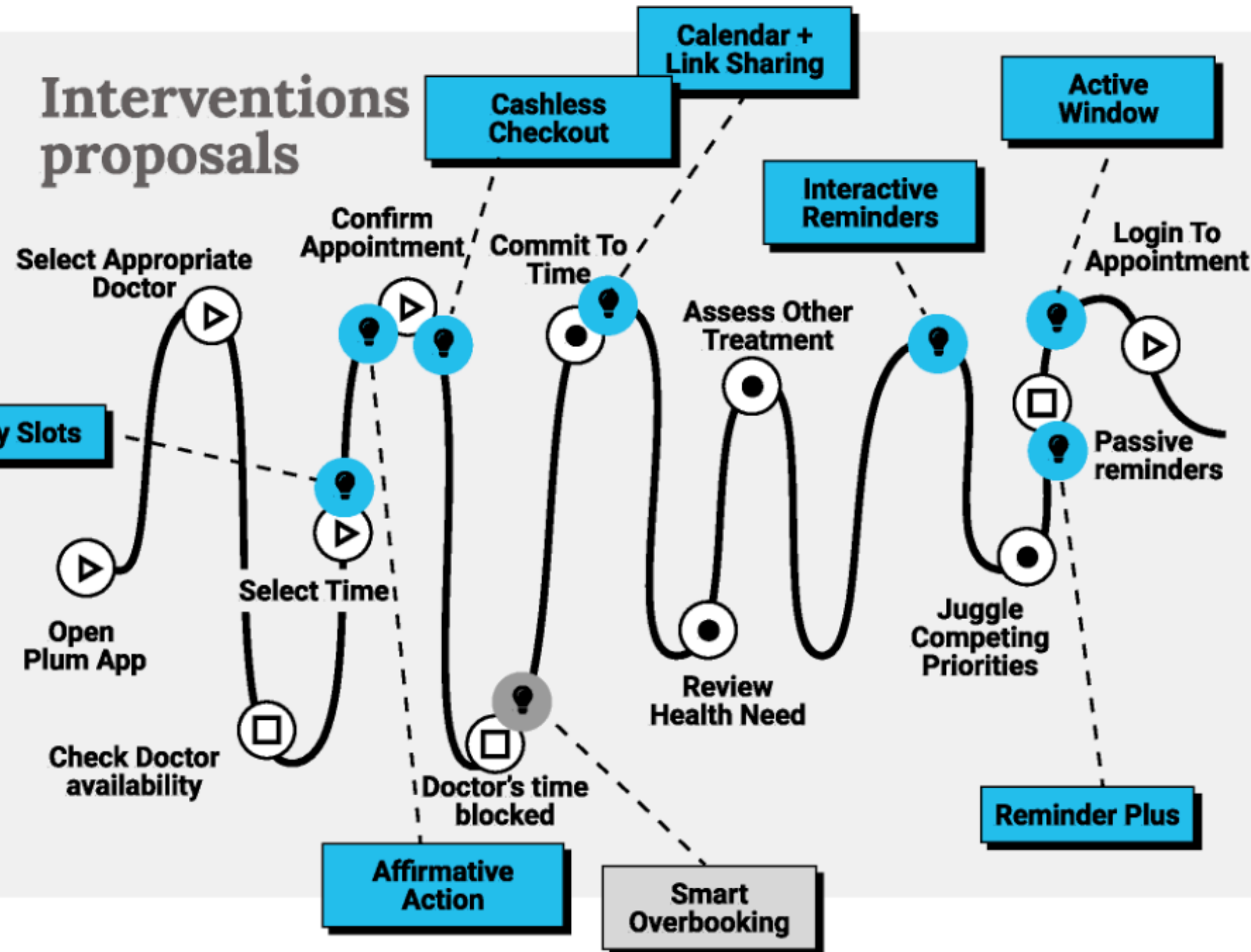
X%
impact

Reach for RICE score based on barriers targeted and % impact

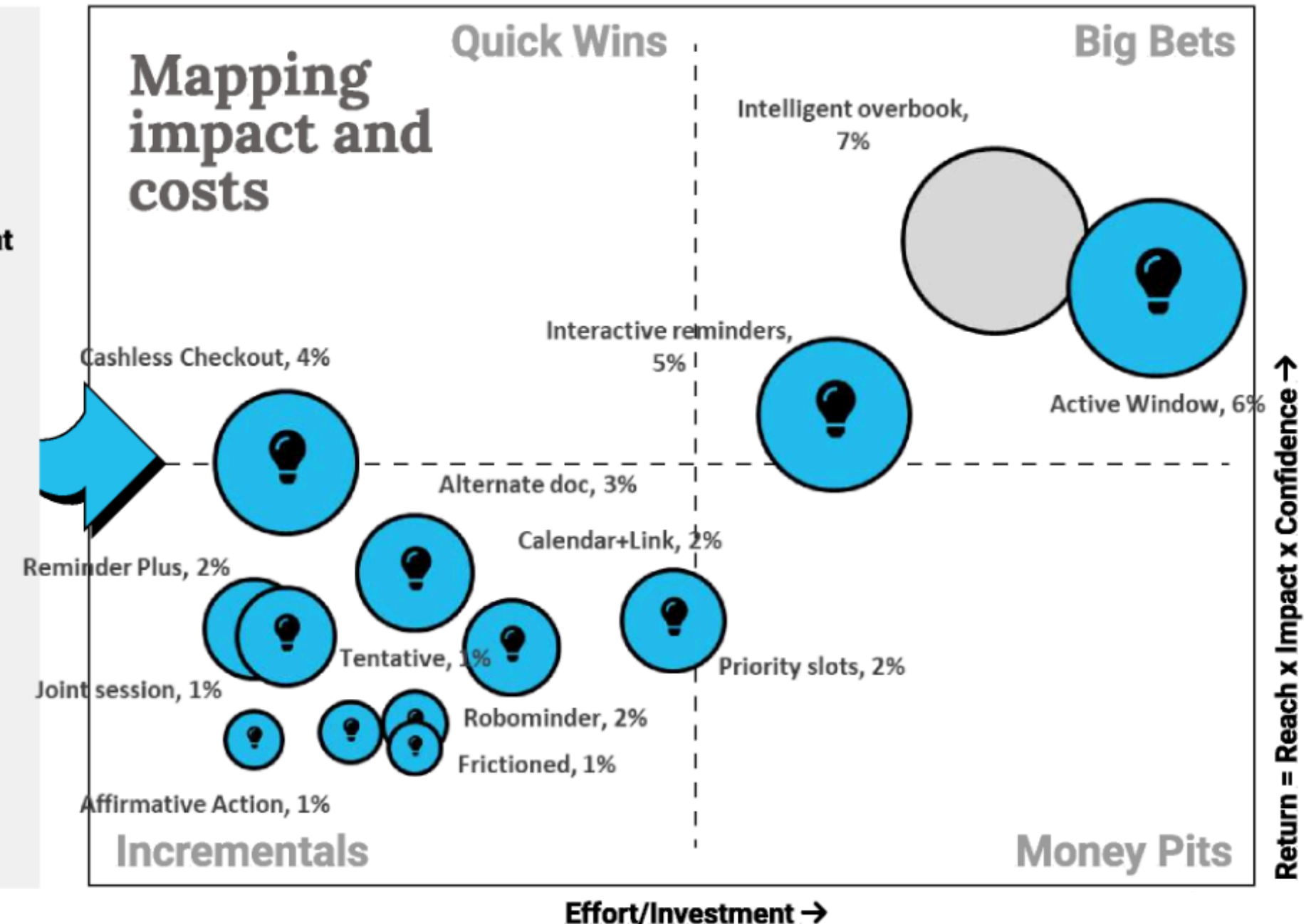


Estimates of individual reduction in no-shows/late cancels are heuristic guesses

Interventions proposals



Mapping impact and costs



Note: Bubble size indicates the % absolute reduction in cancellations + no shows

New Booking Step

Review Your Booking



Dr. Nalanda Debnath Singh

General physician

20 Aug 21 • 2:30 PM

Patient name

Ramani Varun

Summary of Charges

General consultation Rs 500

Platform convenience fee Rs 40

Total Rs 540

You don't have to pay

Wow! This consultation is eligible for cashless settlement and will be charged to your employers account. Simply enter your employee code and click Authorise to charge this.



Re-enter employee code

Authorise

Authorise Payment To Confirm

Your appointment and consultations details are private and never shared with your employer

Intervention Proposal Cashless Checkout

Introduce friction in the booking flow to reframe the free service as a per transaction value, paid for but cashless for employee service funded by the employer

1. Added a new final 'Review and Pay' step
2. The consultation is given a notional value based on market value of comparable services
3. To re-enforce the cashless proposition, the employee is asked to confirm his employee code, and authorise the payment by employer
4. A small delay is introduced while the system authorizes the payment by charging the corporate account
5. Finally the user clicks 'Confirm' button with full knowledge of the service cost

Delay demonstrates evaluation

You don't have to pay

Wow! This consultation is eligible for cashless settlement and will be charged to your employers account. Simply enter your employee code and click Authorise to charge this.



EAB124399

Checking...

Approval with tangible value

You don't have to pay

Wow! This consultation is eligible for cashless settlement and will be charged to your employers account. Simply enter your employee code and click Authorise to charge this.



EAB124399
Employee Name: Varun Siddhi

Approved
Rs 540



Why it will work

“

People have bundling bias; they cannot estimate price of a transaction as long as the price is not tied to a transaction



Pre-mortem Why it will fail



Increased price sensitization friction might reduce service utilization (Need to A/B test variations)



Messaging of “unlimited” redemption might contradict intervention (Use in conjunction with other interventions)

4%

Reduction in no-shows and late cancellations in total



A/B Test

Required sample size per variation = 900*

Expected duration = 7 days

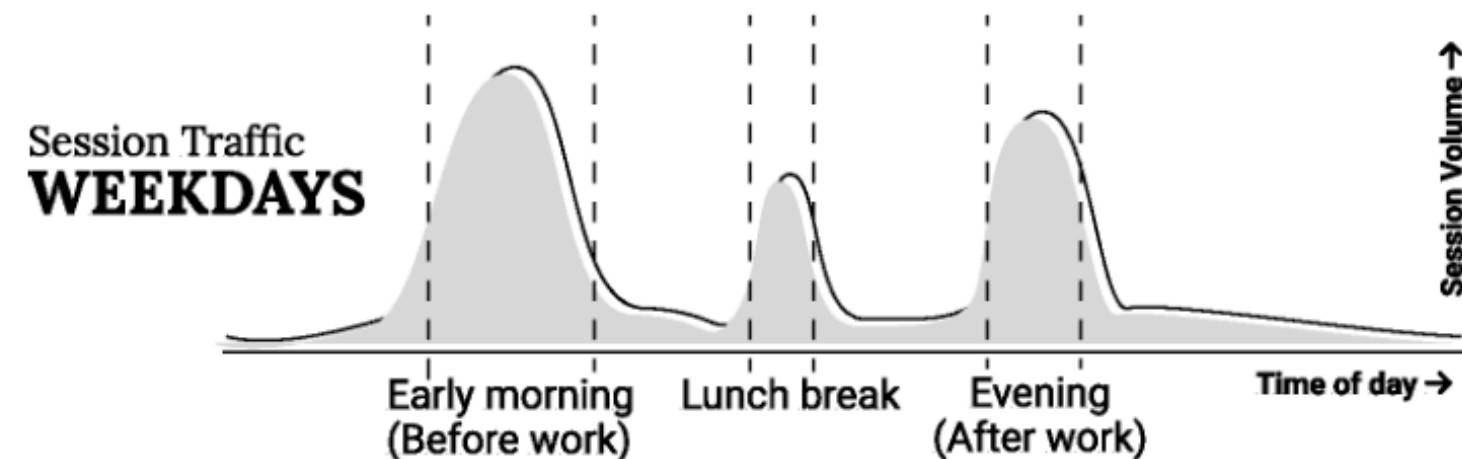
*assuming 90% power

Intervention Proposal

Intelligent Overbooking

An overbooking system which balances the trade-off between increasing **user wait times** (friction) and **improved access to care**. Alongside transparent disclosure the system can **introduce scarcity** (positive friction) while incentivizing users to move to leaner hours.

1. Score signals correlating (ML) to user cancellation and no-shows - history of attendance, lead time, demographics, time of day, ailment type, doctor, engagement with reminders
2. Update expected peak traffic hours based on past data by day of week, to create a traffic score



3. Check for imbalances in availability of rostered doctors, and expected peak hours

12:30 PM 1:30 PM 2:30 PM

3:30 PM 4:30 PM 5:30 PM

6:30 PM 7:30 PM 8:30 PM

Expect A Small Waiting Time ⚡

You've chosen a slot at a very busy time, and the doctors session with a previous patient might lead to small delay (upto 15 mins). It's not cool when you have to wait, but we'll keep you informed in real-time.

Next

Upfront Disclosure

System Principles

- No impact on instant bookings <10 mins
- Minimize impact on users with good attendance history
- Tag overbooking decisions during time selection itself and transparently disclose
- Provide real-time updates on queue (if applicable) to reduce regret



Why it will work



“ Disclosure of busy periods and waiting time can subtly build scarcity



Pre-mortem Why it will fail

- ✗ Due to low capacity utilization (currently), implementing overbooking might be overkill until service scale has been achieved
- ✗ Increased wait times might dilute the on-demand nature of service (evaluate variations in advance disclosure)

7%

Mitigation of capacity loss to no-shows/late cancellations



A/B Test

Required sample size per variation = **300**

Expected duration = **2 days**

*assuming 90% power

Intervention Proposal

Pre-appointment Engagement

Interactive Reminders



2x

Human reminders can be 2x more effective than robocalls, while improving on self-triage errors

Can we be easily **A/b tested** for efficacy without full scale investment

Will drive increase in **proactive cancellations** and attendance rates

Can help better **triage conditions** requiring GP's instead of specialists, helping open up more slots for those who really need it

Prone to high **failure rates** (20-30%) due to connectivity, inaccessibility and increased **operational cost** (targeting)

5% Reduction Expected

5d A/b test window

Reminder Plus

Providing **personalized advice** and **constructive low cost actions** along with the reminder can help people overcome the negative task and mis-estimation barrier

2% Reduction Expected

Re-assert who you are meeting

Remind time left

Build tension around doctors commitment



Your consultation is in 34 mins
Dr. Singh is preparing to meet you on time at 9.45am. Keep your pressure medication history ready.

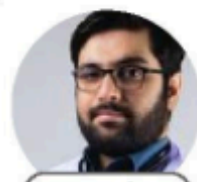
Upload History

Postpone By 30 mins

Constructive action

Low cost cancellation alternative

Build engagement and speed up consultation with detailed bot based complaint capture



34 mins

Preconsultation check

Please check your connectivity and be ready to join at least 5 mins before. If you're running late, you can always release your slot early to help others.

Check My Connection

Release My Slot

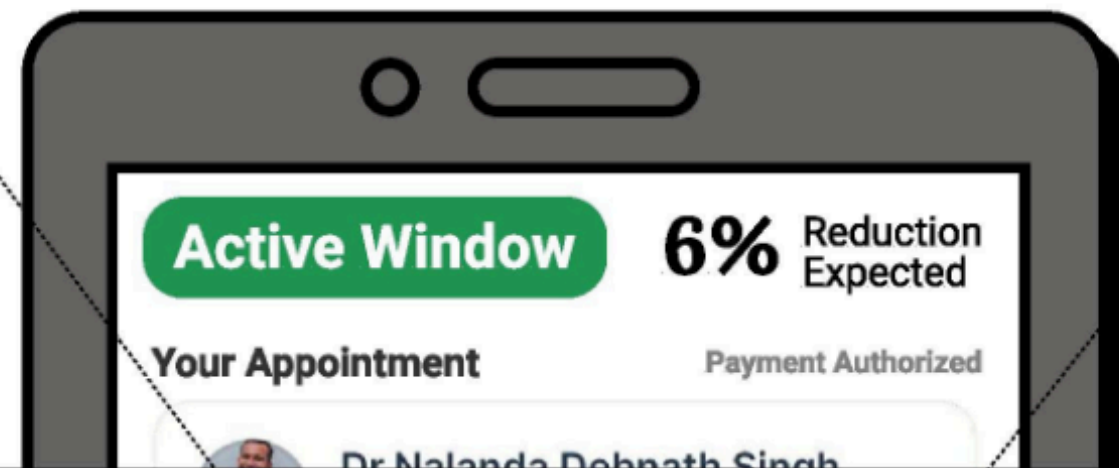
Positive reframing of negative task

Eliminate blank screens for doctors with 5 mins joining buffer

Build commitment with a simulated queue

Build a virtual clinic setting to help users engage and visualize the consultation from 40 mins before

Constructive action



Join call

opens at 9.40am

If you don't join the call before 9.45am, your appointment will be cancelled

Running Late? Cancel early to help 1 person in the queue after you

CANCEL

34 mins no waiting expected



RESCHEDULE

Dr. Singh is seeing patients



online

In a hurry? We'll inform you if an earlier slot opens up

Update your history

Metoprolol X Allergic to Penicillin

Aspirin



Your last consultation with Dr. Srinivas NT was 93 days ago

Update

Provide Details For A Quicker Diagnosis



Sheena

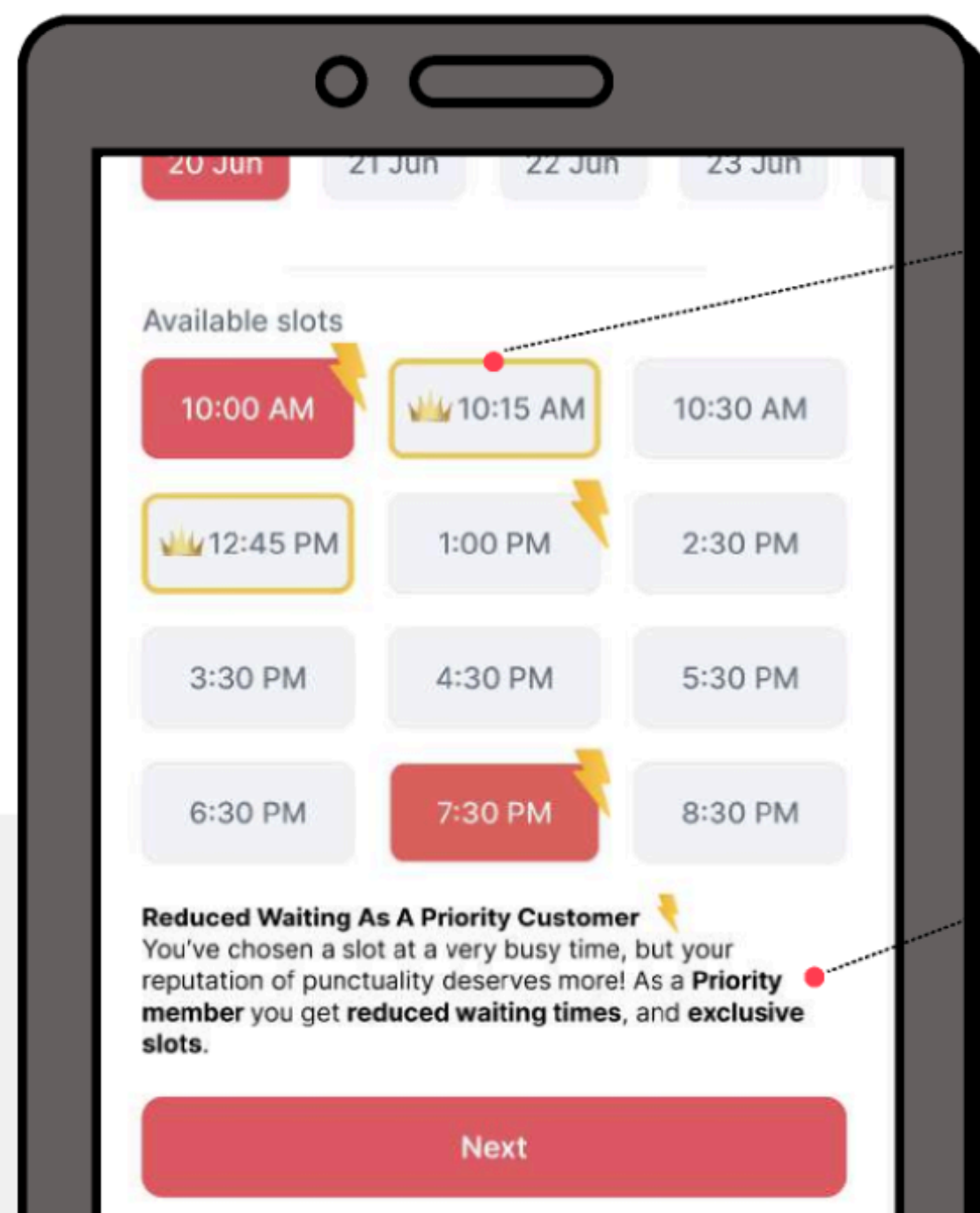
Hi Varun. I'm assisting Dr. Singh today. Could you help me with some details about your complaint?

Intervention Proposal

Priority Service

This a future proofing intervention in the larger scheme of Plum's ambition to become a best-in-class opt-in service at scale.

In a fixed fee or subsidized fee model, creating non-monetary rewards (as opposed to penalties) can help reinforce good behaviour - on-time, or early cancellations



Odd-hour priority slots perk

Reduced wait time as a perk

Priority service users get excluded from overbooking or have more buffer time with previous bookings

Intelligent Overbooking

Frame reminders to promote constructive action to avoid losing priority status

Reminder Plus

Demonstrate value with zero convenience charges for priority users

Cashless Checkout

Active Window

Showcase user as a priority user in the live queue, reduced waiting time



Why it will work

“ People enjoy Endowment when they feel like they've earned a reward, and once enrolled maintaining Status Quo of an elevated service level can be a powerful motivator

Complementing Other Interventions

Priority Service

Feature onboarding

First booking free trial of service

Consultation completed or cancelled

Offer opt-in enrollment with T&C

2%

Reduction in no-shows and late cancellations in total



A/B Test

Required sample size per variation = 3600

Expected duration = 1 month

*assuming 90% power

Next Steps

Build, experiment, rollout

Priority	Intervention	Impact Metric	Reach x Barriers	A/B Experiment Design (90% power)	
P0	 Cashless Checkout Unbundle value of service from fixed company benefits	no-shows -1% -3% late cancel	70%  Forgetting  Low commitment	n = 900 7 days	Variations To Test 1. Pay from company wallet 2. Pay from employee health wallet 3. Validate interest in buying add-on subscription ★
P0	 Reminder Plus Use constructive reframing, blend personalized advice to bring value to reminders	no-shows -1% -1% late cancel	70%  Cancellation Friction  Negative Task Perception  Over-optimistic Planning	n = 3,900 1 month	Variations To Test 1. Postpone CTA 2. Release my slot CTA 3. Cancel CTA
P0	 Calendar Integration + Link Sharing Can help early cancellation by surfacing conflicts with other work; no need for presence of primary	no-shows 0% -2% late cancel	50%  Forgetting  Low commitment	n = 4,300 1 month	Feature Details 1. Add invite to calendar 2. Share video call link to dependent phone for joining without app login
P1	 Active Window Collection of commitment builders ramping up to appointment time, allowing better planning to show up or cancel early	no-shows -2% -4% late cancel	80%  Forgetting  Low commitment  Over-optimistic Planning  Negative Task Perception	n = 400 3 days	Phased Rollout 1. Join buffer (5 mins) before call 2. Live queue 3. Expedite my slot 4. History capture
P2	 Intelligent Overbooking Will not lead to change in user behaviour but will mitigate harm	no-shows -2% -5% late cancel	50%  Forgetting  Low commitment  Over-optimistic Planning  Negative Task Perception  Cancellation Friction	n = 300 2 days	Variations To Test 1. Transparent disclosure at booking 2. Disclosure post booking if applicable

Woah! All this just for me?



P0



Let's make it happen

Image credits: catalyststuff on
freepik, flaticon

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